4

Physics of Matter

CHAPTER OUTLINE

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CENGAGE

4.1 Matter: Phases, Forms, and Forces- Phases of Matter

- **Matter** is anything that has mass and occupies space.
- Obviously, matter exists in different forms that have different physical properties and can be classified into four categories that are called **phases**, or *states*, of matter.

DEFINITION

Solids: Rigid; retain their shape unless distorted by a force

Liquids: Flow readily; conform to the shape of a container; have a well-defined boundary (surface); have higher densities than gases

Gases: Flow readily; conform to the shape of a container; do not have a well-defined surface; can be readily compressed (squeezed into a smaller volume)

Plasmas: Have the properties of gases and also conduct electricity; interact strongly with magnetic fields; commonly exist at higher temperatures

4.1 Matter: Phases, Forms, and Forces- Forms of Matter

- The chemical **elements** represent the simplest and purest forms of everyday matter.
- Each element is composed of incredibly small objects called **atoms**.
- Every **atom** has a very dense, compact core called the **nucleus**.

  The nucleus is composed of two kinds of particles:

  Protons: have a positive electric charge

  Neutrons: have no electric charge
- The nucleus is surrounded by one or more particles called **electrons**.
- Electrons have the same electric charge as protons but are negatively charged.
- Every atom associated with a particular element has the same unique number of protons.

  For example, atoms that have 2 protons are atoms of helium

4.1 Matter: Phases, Forms, and Forces- Forms of Matter

- The ***atomic number*** of an element is the number of protons that are in each atom of the element.

- Each element is also given a brief identifier called its *chemical symbol.*

Element	Symbol	Atomic Number	Phase
Hydrogen	H	1	Gas
Helium	He	2	Gas
Carbon	C	6	Solid
Nitrogen	N	7	Gas
Oxygen	O	8	Gas
Neon	Ne	10	Gas
Sodium	Na	11	Solid
Aluminum	Al	13	Solid
Silicon	Si	14	Solid
Chlorine	Cl	17	Gas
Calcium	Ca	20	Solid
Iron	Fe	26	Solid
Nickel	Ni	28	Solid
Copper	Cu	29	Solid
Zinc	Zn	30	Solid
Silver	Ag	47	Solid
Gold	Au	79	Solid
Mercury	Hg	80	Liquid
Lead	Pb	82	Solid
Uranium	U	92	Solid

4.1 Matter: Phases, Forms, and Forces- Behavior of Atoms and Molecules

- The constituent particles of atoms and molecules exert electrical forces on each other.
- The nature of these forces determines the properties of the substance.
- The forces between atoms in an element depend on the configuration of the electrons in each atom.
- In a compound, the size and shape of the molecules, as well as the forces between the molecules, affect its observed form and properties.

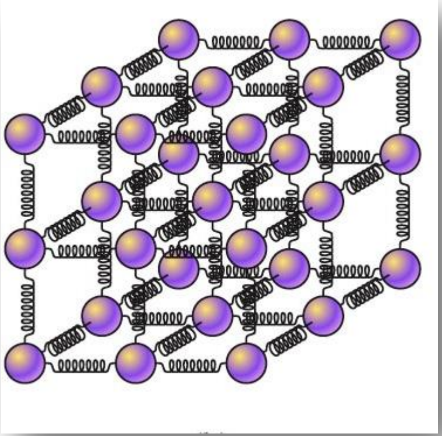
4.1 Matter: Phases, Forms, and Forces- Behavior of Atoms and Molecules

We can relate the three common phases of matter to the interparticle forces as follows:

- **Solids:** Attractive forces between particles are very strong; the atoms or molecules are rigidly bound to their neighbors and can only vibrate.
- **Liquids:** The particles are bound together, though not rigidly; each atom or molecule move about relative to the others but is always in contact with other atoms or molecules.
- **Gases:** Attractive forces between particles are too weak to bind them together; atoms or molecules move freely with high speed and are widely separated; particles are in contact only when they collide.

4.1 Matter: Phases, Forms, and Forces- Behavior of Atoms and Molecules

A standard model for representing the interparticle forces in a solid consists of each atom connected to its neighbors by a spring. The atoms are free to oscillate like a mass hanging from a spring.



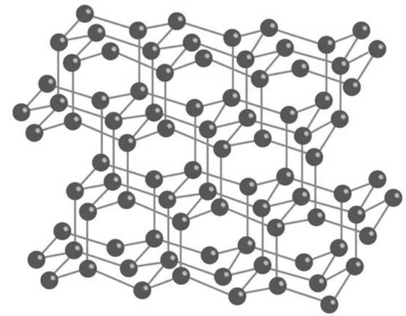
Often atoms or molecules in a solid form a regular geometric pattern called a ***crystal***. Table salt is a crystalline compound in which the sodium and chlorine atoms alternate with each other

In solids that do not have a regular crystal structure, called *amorphous solids*, the atoms or molecules are "piled together" in a random fashion. Glass is a good example of such a solid.

4.1 Matter: Phases, Forms, and Forces- Behavior of Atoms and Molecules

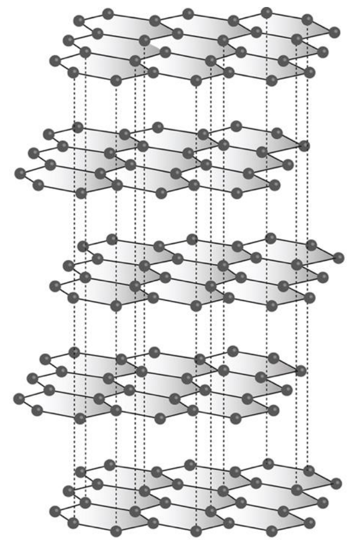
Carbon has two common crystalline forms:

In diamond, each carbon atom is strongly bond to each of its four nearest neighbors resulting in a crystalline solid that is the hardest known natural material.



Diamond

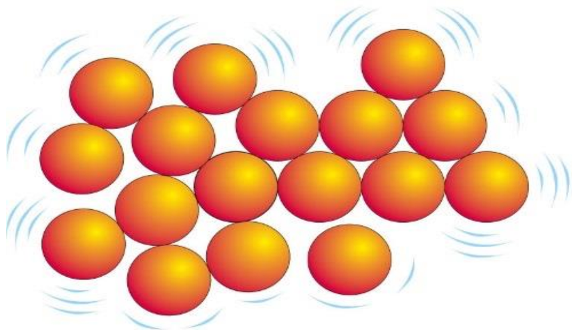
In graphite the carbon atoms form sheets, with each atom strongly bond to its three nearest neighbors in the same layer to form a mesh of hexagons sheets can easily be forced to slide relative to each other, making graphite an excellent “dry” lubricant.



Graphite

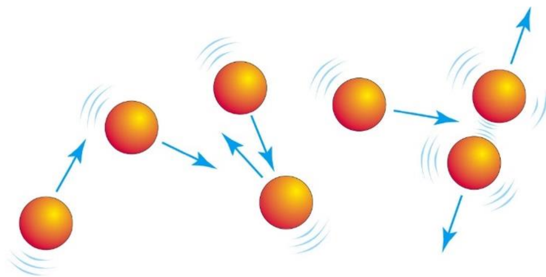
4.1 Matter: Phases, Forms, and Forces- Behavior of Atoms and Molecules

In liquids, the inter-atomic forces are insufficient to bind the atoms rigidly. The atoms are free to move and vibrate.



Liquid

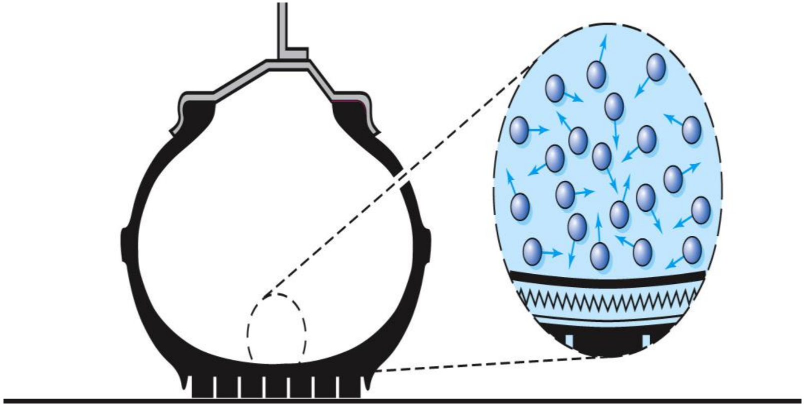
In gases, the atoms and molecules are widely separated and move around Independently, except when they collide, and their speeds are surprisingly high. As each molecule collides randomly with other molecules, its direction of travel is altered, and its speed is sometimes increased and other times decreased.



Gas

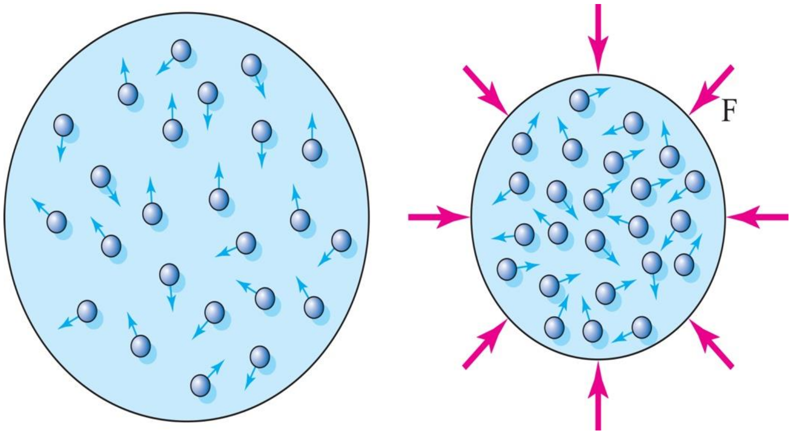
4.1 Matter: Phases, Forms, and Forces- Behavior of Atoms and Molecules

Often, the surface of a solid or a liquid forms a boundary for a gas. The inside of the walls of a tire is a boundary for the air inside. The high-speed atoms or molecules in the gas exert a force on the surface as their random motions cause them to collide with it. These forces are responsible for the pressure that keeps the tire from collapsing. If the molecules are stationary, there is no pressure.



4.1 Matter: Phases, Forms, and Forces- Behavior of Atoms and Molecules

- At normal temperatures and pressures, the average distance between the centers of the atoms or molecules in a gas is about 10 times that in a solid or liquid.
- There is a great deal of empty space in a gas, and that is why gases can be compressed easily.
- If you compress it enough, you force the atoms close enough together that you form a liquid



4.1 Matter: Phases, Forms, and Forces- Behavior of Atoms and Molecules

- The118 different elements—form the basis of matter
- Molecules are composed of two or more atoms “stuck” together to form the basic unit of compounds.
- The nature of the forces between the building blocks (atoms or molecules) determines whether the substance is a solid, a liquid, or a gas.
- Mixtures and solutions consist of different elements or compounds mixed together.

4.2 Pressure: Defining Pressure

The physical quantities **pressure** and **density**, are extensions of the concepts of force and mass, respectively.
They are two essential quantities for studying fluids both at rest and in motion.

DEFINITION

Pressure: The force per unit area when the force acts perpendicular to a surface. The perpendicular component of a force acting on a surface divided by the area of the surface.

$$p = \frac{F}{A}$$

Pressure is a scalar; there is no direction associated with it.

4.2 Pressure: Defining Pressure

The following table lists the common pressure units of measure, in both the *metric system* and the *English system*, and their abbreviations.

Physical Quantity	Metric Units	English Units
Pressure (p)	pascal (Pa) (1 Pa = 1 N/m^2)	pound per square foot (lb/ft^2)
	erg	pound per square inch (psi)
	millimeters of mercury (mm Hg)	inches of mercury (in. Hg)

One **atmospheric pressure** is the average pressure exerted by air at sea level

$$1 \text{ atm} = 1.01 \times 10^5 \text{ Pa} = 14.7 \text{ psi}$$

Example 4.1

A 160-pound person stands on the floor. The area of each shoe that is in contact with the floor is 20 square inches. What is the pressure on the floor?

$$P = \frac{F}{A} = \frac{80 \text{ lb}}{20 \text{ in.}^2} = 4 \text{ psi} \left(\text{standing on both feet}\right)$$

$$P = \frac{160 \text{ lb}}{20 \text{ in.}^2} = 8 \text{ psi} \left(\text{standing on one foot}\right)$$

Example 4.1, continued

- What if the person put on high-heeled shoes and balanced on the heel of one shoe? The bottom of the heel measures 0.5 inches by
- 0.5 inches.

$$P = \frac{160 \text{ lb}}{0.25 \text{ in.}^2} = 640 \text{ psi}$$

$$\left(\text{balanced on a narrow heel}\right)$$

Example 4.2

The cabin pressure of a passenger jet cruising at high altitude (25,000 ft) is about 6 psi (0.41 atm) greater than the pressure outside. What is the outward force on a window measuring 1 foot by 1 foot and on a door measuring 1 meter by 2 meters?

Window:

$$A = 1\text{ ft} \times 1\text{ ft} = 12\text{ in.} \times 12.\text{in.} = 144\text{ in.}^2$$

$$F = pA = 6\text{ psi} \times 144\text{ in.}^2 = 864\text{ lb}$$

For the door, we use SI units:

$$p = 6\text{ psi} = 6 \times 1\text{ psi} = 6 \times 6{,}890\text{ Pa} = 41{,}340\text{ Pa}$$

$$A = 1\text{ m} \times 2\text{ m} = 2\text{ m}^2$$

Consequently, the force on the door is

$$F = pA = 41{,}340\text{ Pa} \times 2\text{ m}^2$$

$$= 82{,}680\text{ N}$$

4.2 Pressure: Gauge Pressure

Pressure is a relative quantity. This means, when you measure a pressure, you are measuring it relative to some other pressure.

When you measure your tire pressure, you are measuring the pressure in the tire *above* the atmospheric pressure.

When a tire is flat, there is still air inside it, but the pressure inside is the same as the atmospheric pressure outside.

4.2 Pressure: Gauge Pressure

For example, let us say that the atmospheric pressure is 14 psi. A tire is tested, and the air pressure gauge shows 30 psi.

This means that the air pressure inside the tire is 30 psi higher than the air pressure outside the tire. So, the actual pressure on the inner walls of the tire is 30 + 14 = 44 psi. This is sometimes referred to as the *absolute pressure*. The pressure relative to the outside air (30 psi in this case) is then called the *gauge pressure*.

4.2 Pressure: Gauge Pressure

- The **gauge pressure** is the pressure relative to the current atmospheric pressure.

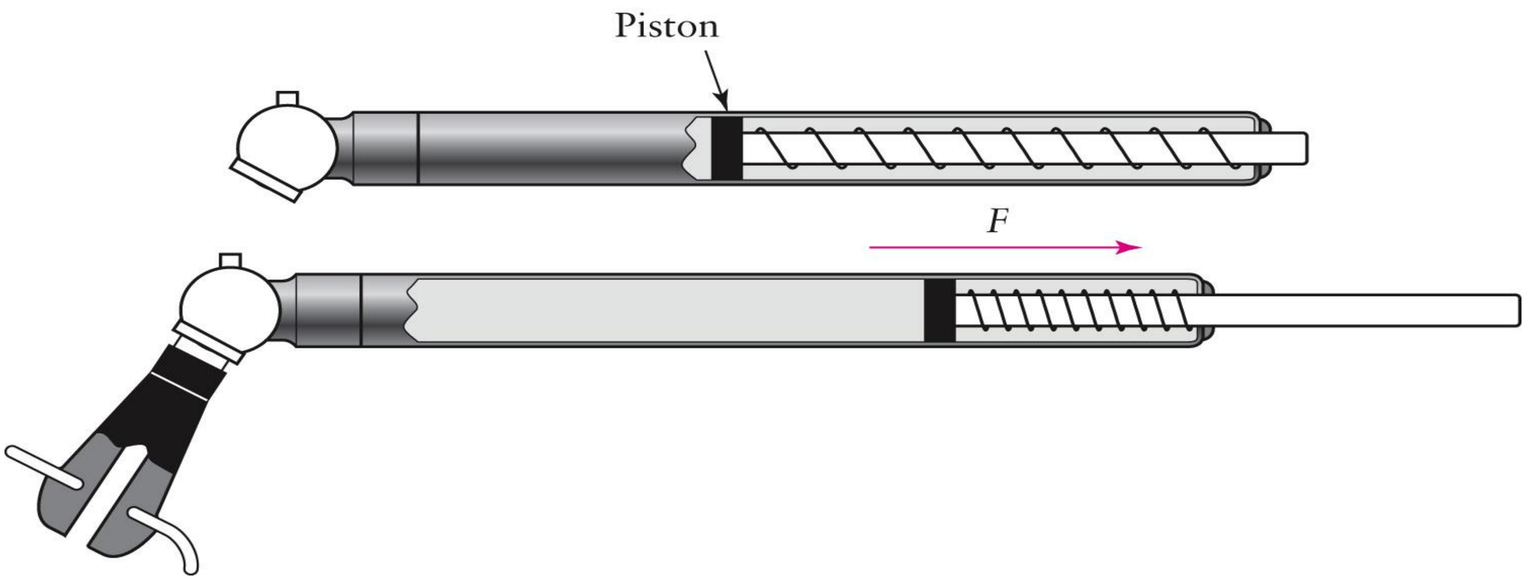
- The **absolute pressure** is the gauge pressure *plus* the atmospheric pressure.

$$p_{\text{absolute}} = p_{\text{gauge}} + p_{\text{atmospheric}}$$

4.2 Pressure: Gauge Pressure

The standard pen-shaped tire gauge compares the tire pressure against a spring and the atmospheric pressure.

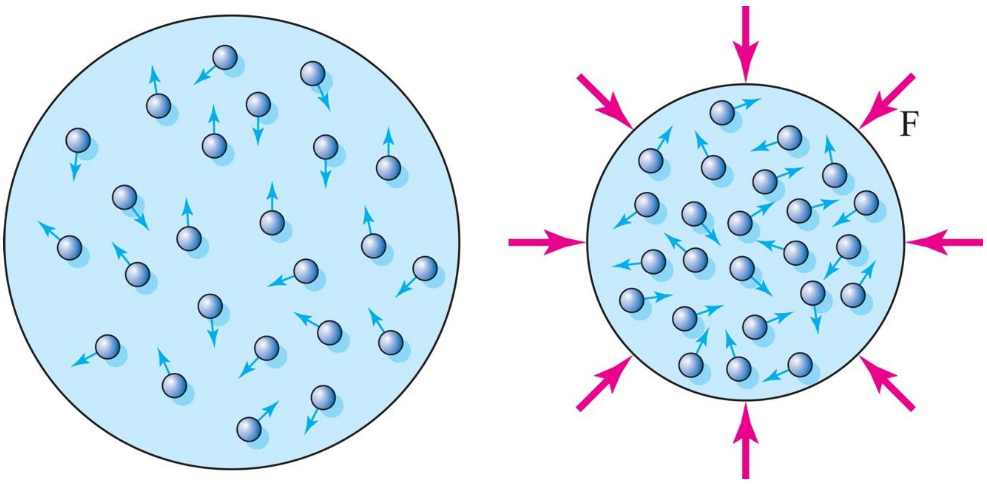
The higher pressure of the air in the tire pushes the piston to the right and compresses the spring. The higher the pressure, the greater the force and the greater the compression of the spring.



4.2 Pressure: Gauge Pressure

Gases are compressible, the volume of a gas can be changed Whenever the volume of a fixed amount of gas is changed, the pressure in the gas changes also.

Increasing the volume of a gas reduces the pressure.
Decreasing the volume increases the pressure.



4.2 Pressure: Gauge Pressure

The temperature of a gas influences the speeds of the atoms or molecules. Consequently, the pressure is also affected by the gas temperature. Increasing the temperature increases the motion of the gas atoms and therefore the pressure.

When the temperature of a given quantity of gas is kept constant

$$pV = \text{ a constant } \quad (\text{gas at fixed temperature})$$

This means that the *volume of a gas is inversely proportional to the pressure*. If the pressure is doubled, the volume is halved.

4.3 Density: Mass Density

Pressure is an extension of the idea of force. Similarly, **mass density** is an extension of the concept of mass. Just as pressure is a measure of the concentration of force, density is a measure of the concentration of mass.

DEFINITION

Mass Density: The mass per unit volume of a substance. The mass of a quantity of a substance divided by the volume it occupies.

$$D = \frac{m}{V}$$

Density is a scalar quantity.

4.3 Density: Mass Density

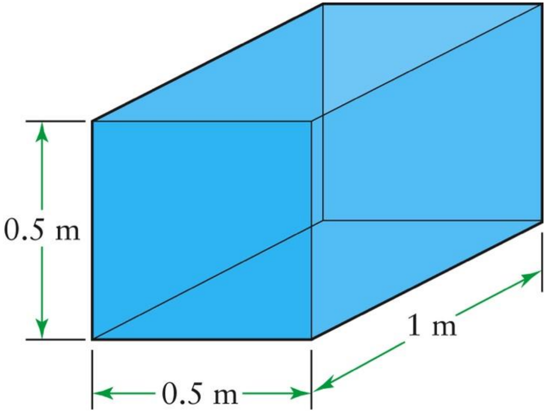
The following table lists the common Mass density units of measure, in both the *metric system* and the *English system*, and their abbreviations.

Physical Quantity	**Metric Units**	**English Units**
Mass density (*D*)	kilogram per cubic meter (kg/m^3)	slug per cubic foot
	gram per cubic centimeter (g/cm^3)	

To find the mass density of a substance, one measures the mass of a representative part or "sample" of it and divides by the volume of that part or sample. It doesn't matter how much is used: the greater the volume, the greater the mass.

Example 4.3

The dimensions of a rectangular aquarium are 0.5 meters by 1 meter by 0.5 meters. The mass of the aquarium is 250 kilograms larger when it is full of water than when it is empty. What is the density of the water?



$$\mathrm{V} = \mathrm{l} \times \mathrm{w} \times \mathrm{h}$$

$$= 1\, m \times 0.5\ \mathrm{m}\ \times 0.5\ \mathrm{m}$$

$$= 0.25\, m^3$$

$$D = \frac{m}{V}$$

$$= \frac{250\, kg}{0.25\, m^3} = 1000\, k\, g/m^3$$ (Mass density of water)

Example 4.3, continued

When the same tank is filled with gasoline, the mass of the gasoline is found to be 170 kilograms. What is the density of gasoline?

$$D = \frac{170 \text{ kg}}{0.25 \text{ m}^3} = 680 \text{ kg/m}^3$$

4.3 Density: Mass Density

- The mass density of each element or compound is fixed.
- Water, lead, mercury, salt, oxygen, gold, and so on, all have unique mass densities that have been measured and cataloged.
- The density of a mixture containing two or more substances depends on the density and percentage of each component

4.3 Density: Mass Density

Substance	Type	Mass Density, D (kg/m^3)	Weight Density, D_W (lb/ft^3)	Specific Gravity
Solids				
Styrofoam	m	37	2.3	0.087
Juniper wood	m	560	35	0.56
Ice	c	917	57.2	0.917
Ebony wood	m	1,200	75	1.2
Concrete	m	2,500	156	2.5
Aluminum	e	2,700	168	2.7
Granite	m	2,700	158	2.7
Diamond	e	3,400	210	3.4
Iron	e	7,860	490	7.86
Brass	m	8,500	530	8.5
Nickel	e	8,900	555	8.9
Copper	e	8,930	557	8.93
Silver	e	10,500	655	10.5
Lead	e	11,340	708	11.34
Uranium	e	19,000	1,190	19
Gold	e	19,300	1,200	19.3

Substance	Type	Mass Density, D (kg/m^3)	Weight Density, D_W (lb/ft^3)	Specific Gravity
Liquids				
Gasoline	m	680	42	0.68
Ethyl alcohol	c	791	49	0.791
Water (pure)	c	1,000	62.4	1.00
Seawater	m	1,030	64.3	1.03
Antifreeze	m	1,100	67	1.1
Sulfuric acid	c	1,830	114	1.83
Mercury	e	13,600	849	13.6
Gases (at 0°C and 1 atm)				
Hydrogen	e	0.09	0.0056	0.00009
Helium	e	0.18	0.011	0.00018
Nitrogen (N_2)	c	1.25	0.078	0.00125
Air	m	1.29	0.08	0.00129
Oxygen (O_2)	c	1.43	0.089	0.00143
Carbon dioxide	c	1.98	0.12	0.00198
Radon	e	10	0.627	0.010

Note: "e" stands for element, "c" for compound, and "m" for mixture.

4.3 Density: Mass Density

Having a list of the densities of common substances is quite useful for three reasons.

First, one can use mass density to help *identify* a substance. For example, one can determine whether or not a gold ring is solid gold by measuring its density and comparing it to the known density of pure gold.

Second, density measurement is used routinely to determine how much of a particular substance is present in a mixture. If you have donated blood to a blood bank, part of the screening include checking to see whether the hemoglobin content of your blood was high enough. This is done by determining whether the blood's density is greater than an accepted minimum value.

4.3 Density: Mass Density

Third, one can *calculate the mass* of something if one knows what its volume is. The mass of a substance equals the volume that it occupies times its mass density.

$$m = V \times D$$

Example 4.4

The mass of water needed to fill a swimming pool can be computed by measuring the volume of the pool. Let's say a pool is going to be built that will be 10 meters wide, 20 meters long, and 3 meters deep. How much water will it hold?

$$V = l \times w \times h$$

$$= 20\ m \times 10\ m\ \times 3\ m = 600\ m^3$$

$$m = V \times D = 600\ m^3\ \times 1000\ kg/m^3$$

$$= 600{,}000\ kg$$

4.3 Density: Weight Density and Specific Gravity

In some cases, it is practical to use another type of density called **weight density**.

DEFINITION

Weight Density: The weight per unit volume of a substance. The weight of a quantity of a substance divided by the volume it occupies.

$$D_W = \frac{W}{V}$$

4.3 Density: Weight Density and Specific Gravity

The following table lists the common weight density units of measure, in both the *metric system* and the *English system*, and their abbreviations.

Physical Quantity	**Metric Units**	**English Units**
Weight density D_W	newton per cubic meter (N/m^3)	pound per cubic foot (lb/ft^3)
		pound per cubic inch ($lb/in.^3$)

The weight density of a substance is equal to the mass density times the acceleration from gravity. This is because the weight of a substance is just g times its mass.

$$W = m \times g \rightarrow D_W = D \times g$$

Weight Density

Substance	Type	Mass Density, D (kg/m^3)	Weight Density, D_W (lb/ft^3)	Specific Gravity
Solids				
Styrofoam	m	37	2.3	0.087
Juniper wood	m	560	35	0.56
Ice	c	917	57.2	0.917
Ebony wood	m	1,200	75	1.2
Concrete	m	2,500	156	2.5
Aluminum	e	2,700	168	2.7
Granite	m	2,700	158	2.7
Diamond	e	3,400	210	3.4
Iron	e	7,860	490	7.86
Brass	m	8,500	530	8.5
Nickel	e	8,900	555	8.9
Copper	e	8,930	557	8.93
Silver	e	10,500	655	10.5
Lead	e	11,340	708	11.34
Uranium	e	19,000	1,190	19
Gold	e	19,300	1,200	19.3

Substance	Type	Mass Density, D (kg/m^3)	Weight Density, D_W (lb/ft^3)	Specific Gravity
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Gasoline	m	680	42	0.68
Ethyl alcohol	c	791	49	0.791
Water (pure)	c	1,000	62.4	1.00
Seawater	m	1,030	64.3	1.03
Antifreeze	m	1,100	67	1.1
Sulfuric acid	c	1,830	114	1.83
Mercury	e	13,600	849	13.6
Gases (at 0°C and 1 atm)				
Hydrogen	e	0.09	0.0056	0.00009
Helium	e	0.18	0.011	0.00018
Nitrogen (N_2)	c	1.25	0.078	0.00125
Air	m	1.29	0.08	0.00129
Oxygen (O_2)	c	1.43	0.089	0.00143
Carbon dioxide	c	1.98	0.12	0.00198
Radon	e	10	0.627	0.010

Note: “e” stands for element, “c” for compound, and “m” for mixture.

Example 4.5

A college dormitory room measures 12 feet wide by 16 feet long by 8 feet high. What is the weight of air in it under normal conditions?

$$W = D_W \times V$$

$$V = l \times w \times h = 16 \text{ ft} \times 12 \text{ ft} \times 8 \text{ ft} = 1{,}536 \text{ ft}^3$$

$$W = D_W \times V = 0.08 \text{ lb/ft}^3 \times 1{,}536 \text{ ft}^3 = 123 \text{ lb}$$

4.3 Density: Weight Density and Specific Gravity

- **The specific gravity** of a substance is the ratio of its density to the density of water.
- Water is used as the reference for specific gravities simply because it is such an important and yet common substance in our world.
- The density of water is 1,000 kg/m^3

$$SP = \frac{D_{Substance}}{D_{Water}}$$

- It is a ratio for the fraction of submerged object.
- Example: D(ice) = (920 kg/m^3)

SP=0.92

92% of it is under water

4.4 The Law of Fluid Pressure

- We live in a sea of air—the atmosphere that exerts pressure on everything in it.
- This pressure varies with altitude and is caused by the force of gravity.
- There are two general properties of pressures in fluids.

  first, fluid pressure act in all directions.

  second, the force of gravity causes the pressure in a fluid to vary with depth only, not with horizontal position.

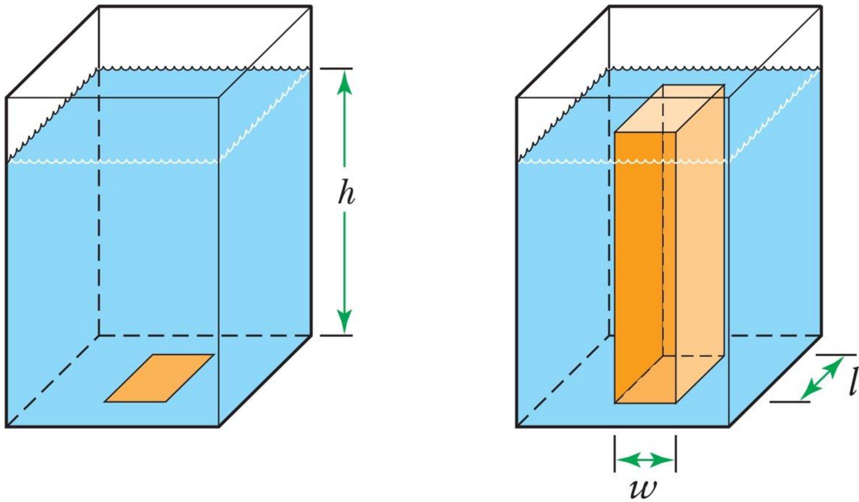
4.4 The Law of Fluid Pressure

LAWS

Law of Fluid Pressure: The (gauge) pressure at any depth in a fluid at rest equals the weight of the fluid in a column extending from that depth to the "top" of the fluid divided by the cross-sectional area of the column.

- This law explains how the pressure in a fluid is related to gravity.
- This law is as much a prescription for determining the pressure in a fluid as it is a description of what causes pressure in a fluid

4.4 The Law of Fluid Pressure



A tank is filled with a liquid to a depth h. Any rectangular area on the bottom supports the weight of all of the fluid directly above it. So the pressure on the bottom equals the weight of the liquid in the column divided by the area of that rectangle.

$$P = \frac{F}{A} = \frac{W}{A} = \frac{\text{weight of liquid}}{\text{area of rectangle}}$$

It does not matter what the actual area is: a larger rectangle will have a proportionally larger amount of liquid in the column above.

The actual height of the column of liquid is what determines the pressure.

4.4 The Law of Fluid Pressure

We compute the pressure using the fact that the weight of the liquid equals the weight density *DW* of the liquid times the volume *V* of the column.

$$F = W = D_W \times V = D_W \times l \times w \times h$$

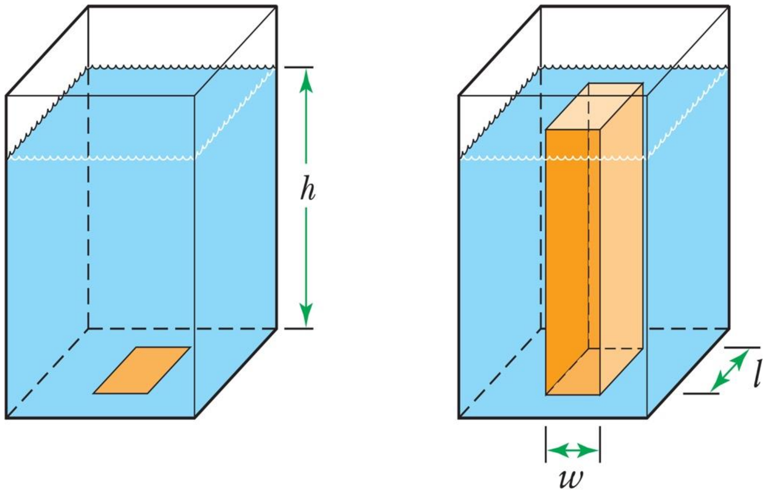
$$A = \text{area of rectangle} = l \times w$$

$$p = \frac{W}{A} = \frac{D_W \times l \times w \times h}{l \times w}$$

$$p = D_W h$$

$$D_W = D_g \rightarrow p = Dgh$$

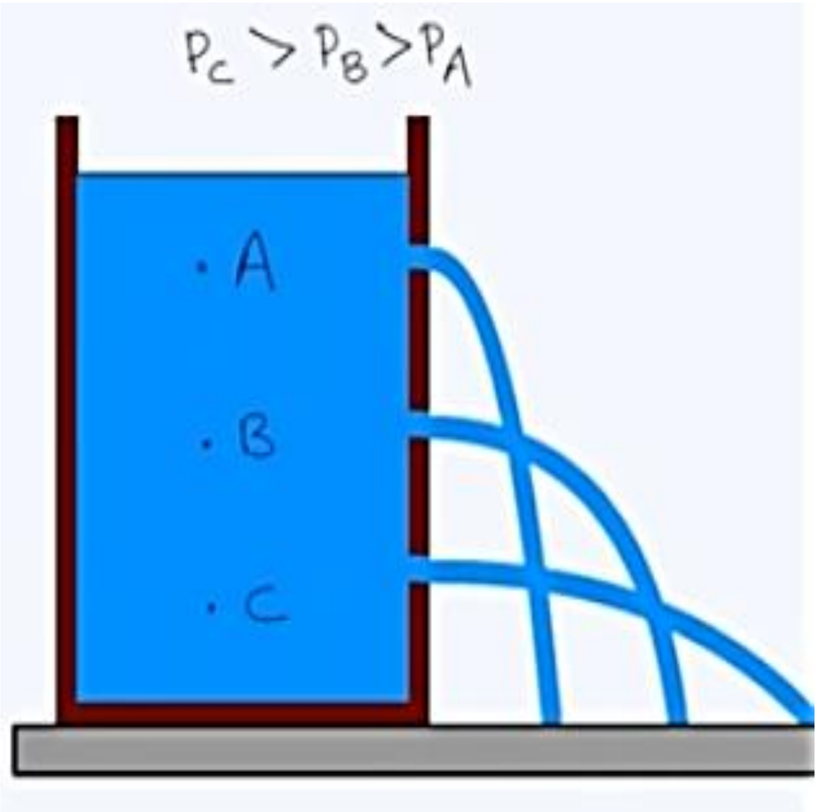
$$(\text{gauge pressure in a liquid})$$



4.4 The Law of Fluid Pressure

- Note that this is the gauge pressure at the depth *h*. To get the absolute pressure, we need to add the atmospheric pressure pushing down at the top of the fluid column.

- In a liquid, the absolute pressure at a depth *h* is greater than the pressure at the surface by an amount equal to the weight density of the liquid times the depth.



Example 4.6

Let's calculate the gauge pressure at the bottom of a typical swimming pool—one that is 10 feet (3.05 meters) deep.

$$p = D_W h = 62.4 \text{ lb/ft}^3 \times 10 \text{ ft} = 624 \text{ lb/ft}^3$$

$$p = 624 \frac{\text{lb}}{\text{ft}^2} = 624 \frac{\text{lb}}{144 \text{ in.}^2} = 4.33 \text{ lb/in.}^2$$

$$= 4.33 \text{ psi} \ (\text{gauge pressure})$$

Absolute pressure = 4.33 psi + 14.7 psi = 19.03 psi

SI units:

$$p = Dgh = 1{,}000 \text{ kg/m}^3 \times 9.8 \text{ m/s}^2 \times 3.05 \text{ m} = 29{,}900 \text{ Pa}$$

4.4 The Law of Fluid Pressure

The general result for the increase in pressure with depth in water is

$$p = 0.433\, psi/ft \times h \qquad \text{(for water, } h \text{ in } ft\text{, } p \text{ in } psi\text{)}$$

Example 4.7

At what depth in pure water is the gauge pressure 1 atm?

$$p = 0.433 \text{ psi/ft} \times h$$

$$14.7 \text{ psi} = 0.433 \text{ psi/ft} \times h$$

$$\frac{14.7 \text{ psi}}{0.433 \text{ psi/ft}} = h$$

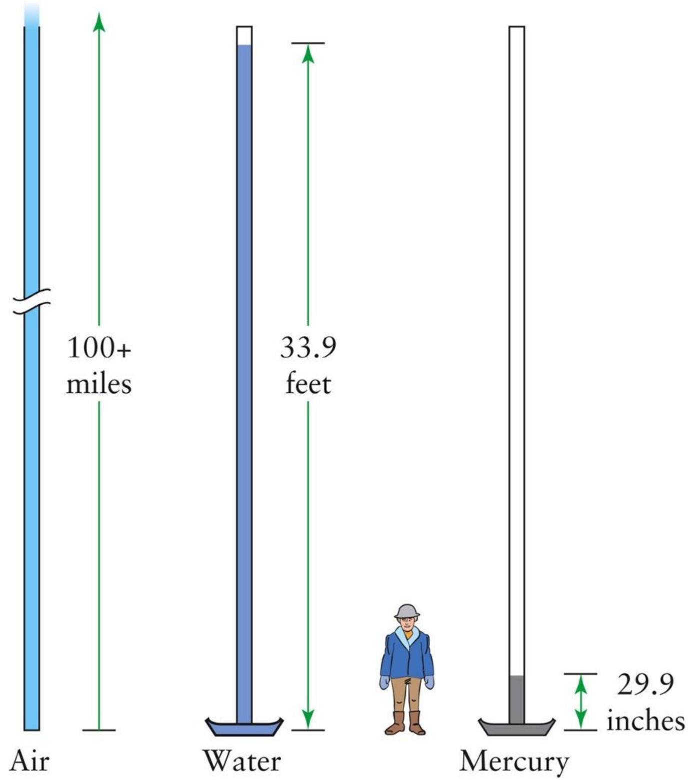
$$h = 33.9 \text{ ft} = 10.3 \text{ m}$$

$$\text{mercury: } h = \frac{33.9 \text{ ft}}{13.6} = 2.49 \text{ ft} = 29.9 \text{ in.} = 0.76 \text{ m}$$

Example 4.7

- The atmosphere exerts a pressure on everything since there is a very high column of air above Earth's surface.

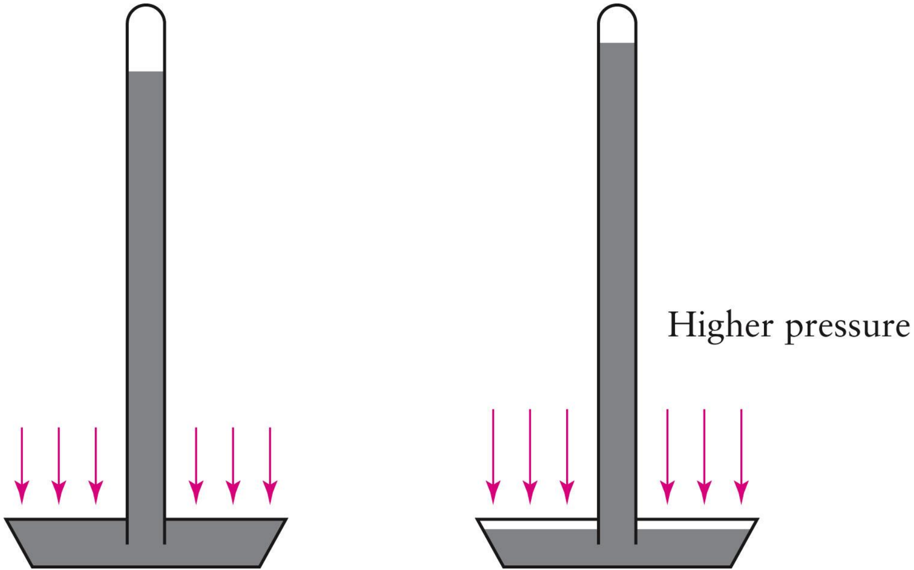
- The column height for air, water and mercury are different because the mass densities are increasingly larger.



4.4 The Law of Fluid Pressure

A device to measure air pressure is a barometer.

- As the air pressure increases, it exerts a larger pressure on the fluid's surface.
- This forces the fluid farther up the tube.
- The height of the column in the tube indicates the air pressure.



4.5 Archimedes' Principle: Buoyancy

- We've found that the force of gravity on a fluid causes the pressure to increase with depth.
- This produces something we call a ***buoyant force*** that exerts an upward force on an object placed in a fluid.

DEFINITION

Buoyant Force: The upward force exerted by a fluid on a substance partly or completely immersed in it.

The buoyant force depends on the density of the fluid and the substance.

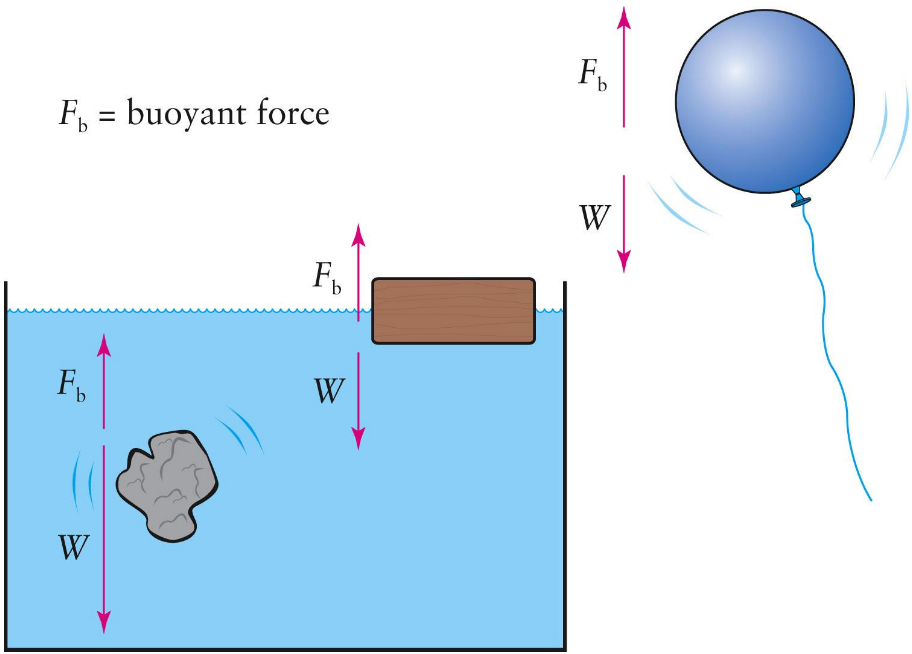
4.5 Archimedes' Principle: Buoyancy

As long as no other forces are acting on the substance, there are three possibilities:

I. If the buoyant force is *less than* the weight of the substance, it will *sink*.

II. If the buoyant force is *equal* to the weight, the substance will *float*.

III. If the buoyant force is *greater than* the weight of the substance, it will *rise* upward.



4.5 Archimedes' Principle: Archimedes' Principle

The size of the buoyant force is determined by Archimedes' Principle

PRINCIPLES

Archimedes' Principle: The buoyant force acting on a substance in a fluid at rest is equal to the weight of the fluid displaced by the substance.

$$F_B = \text{weight of displaced fluid}$$

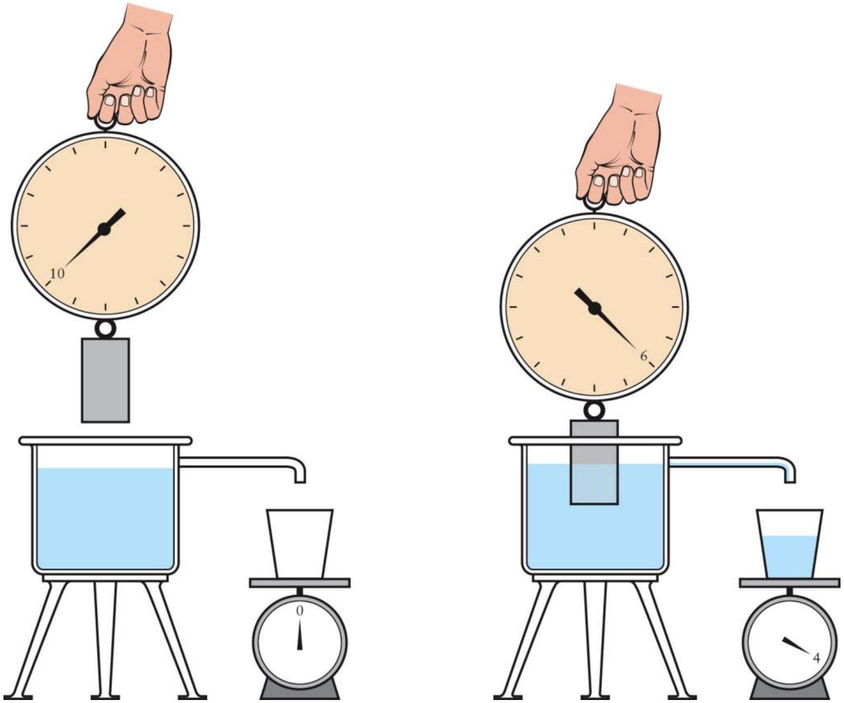
This means the object will sink until the weight of the object equals the weight of the water displaced by the object.

4.5 Archimedes' Principle: Archimedes' Principle

If the weight of an object is 10 newtons.

When the object is immersed in water, the scale reads a smaller force because of the upward buoyant force.

The buoyant force is equal to the weight of the water that the object displaces.



$$\text{scale reading} = \text{weight} - \text{buoyant force}$$

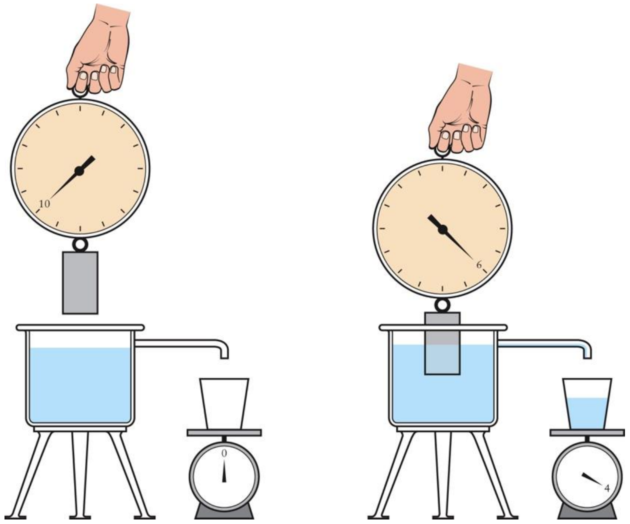
$$6\ \text{N} = 10\ \text{N} - 4\ \text{N}$$

4.5 Archimedes' Principle: Archimedes' Principle

When the scale shows 6 newtons, the buoyant force is 4 newtons.

The weight of the water that spills over the side of the beaker is also 4 newtons.

$$\text{scale reading} = \text{weight} - \text{buoyant force}$$
$$6\text{ N} = 10\text{ N} - 4\text{ N}$$



If the object is lowered farther into the water, the buoyant force will increase, and the scale reading will decrease.

4.5 Archimedes' Principle: Archimedes' Principle

Notice that the buoyant force acting on a substance doesn't depend on what the substance is, only on how much fluid it displaces when immersed.

Identical balloons filled to the same size with helium, air, and water all have the same buoyant force acting on them. Only the helium-filled balloon floats in air because its weight is less than the buoyant force.

Also, the weight of a substance alone does not determine whether or not it will float. A tiny pebble will sink in water, but a 2-ton log will float.

4.5 Archimedes' Principle: Archimedes' Principle

If solid object submerged in a liquid. The volume of the liquid displaced by the object is, of course, equal to the object's volume. So,

$$weight\ of\ object = weight\ density\ (of\ object) \times volume$$

$$W = D_w(object) \times V$$

$$F_B = D_w(fluid) \times V(displaced\ fluid)$$

$$buoyant\ force = weight\ of displaced\ fluid$$

$$F_B = weight\ density\ (of\ fluid) \times volume$$

$$F_B = D_w(fluid) \times V$$

4.5 Archimedes' Principle: Archimedes' Principle

From this, we conclude that if the weight density of the object is greater than the weight density of the fluid, the object's weight is greater than the buoyant force. The object sinks. If its density is less than the fluid's density, it floats.

Example 4.8

A contemporary Huckleberry Finn wants to construct a raft by attaching empty plastic 1-gallon milk jugs to the bottom of a sheet of plywood. The raft and passengers will have a total weight of 300 pounds. How many jugs are required to keep the raft afloat on water?

$$\begin{aligned} F_b &= \text{weight of water displaced} \\ &= D_W(\text{water}) \times \text{volume of water displaced} \end{aligned}$$

$$300\ \text{lb} = 62.4\ \text{lb/ft}^3 \times V \rightarrow V = \frac{300\ \text{lb}}{62.4\ \text{lb/ft}^3} = 4.8\ \text{ft}^3$$

$$4.8\ \text{ft}^3 \times 7.48\ \text{gallons per ft}^3 \approx 36\ \text{jugs}$$

Example 4.9

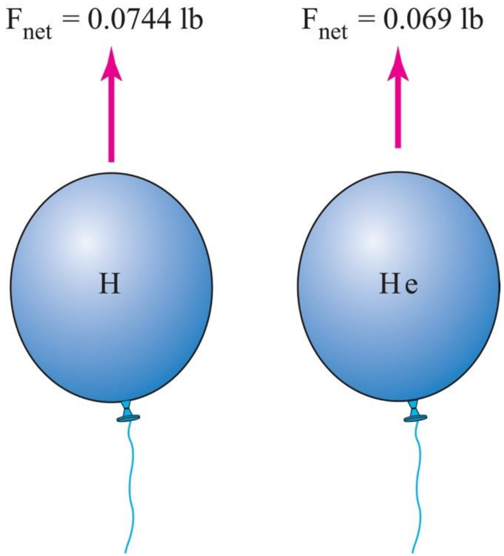
Before the spectacular and tragic destruction of the German airship *Hindenburg* on May 6, 1937, blimps, zeppelins, and balloons were filled with hydrogen. Now helium is used. Let's compare the two gases in terms of their buoyancy in air. Each cubic foot of hydrogen gas weighs 0.0056 pounds at 0 °C and 1 atm.

$$\text{net force} = 0.08\ \text{lb} - 0.0056\ \text{lb}$$

$$F = 0.0744\ \text{lb}\ (\text{hydrogen})$$

$$\text{net force} = 0.08\ \text{lb} - 0.011\ \text{lb}$$

$$F = 0.069\ \text{lb}\ (\text{helium})$$



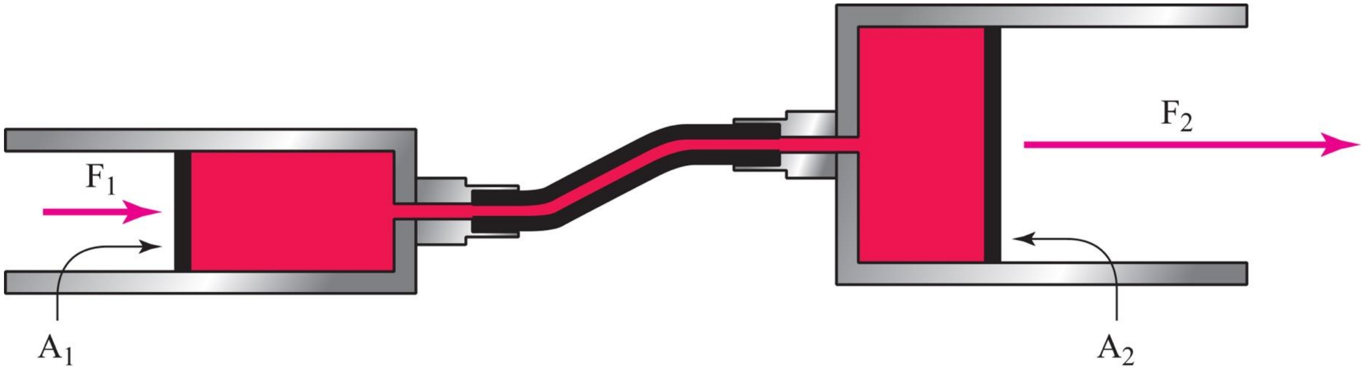
4.6 Pascal's Principle

PRINCIPLES

Pascal's Principle: Pressure applied to an enclosed fluid is transmitted undiminished to all parts of the fluid and to the walls of the container.

This principle is widely used in hydraulic systems.

4.6 Pascal's Principle



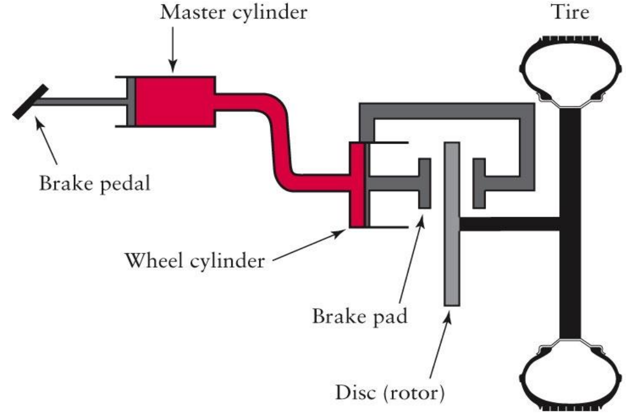
- The pressure caused by the force on the small piston is transmitted throughout the fluid and acts on the larger piston.

- The resulting force on this piston is larger than the force on the smaller piston.

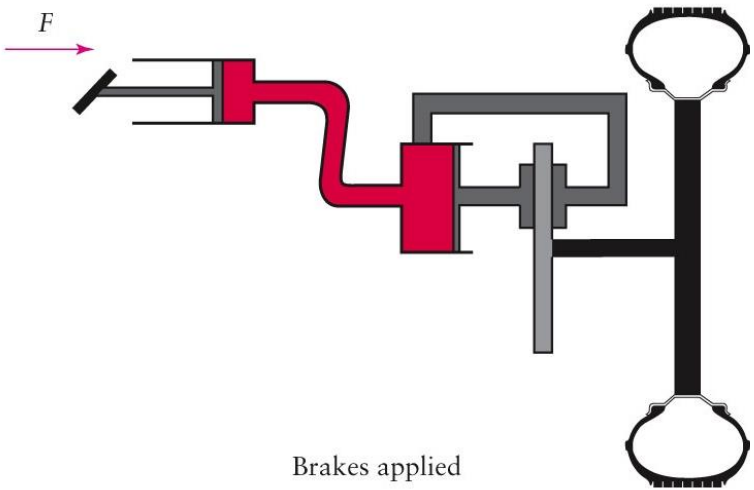
$$\frac{F_1}{A_1} = p = \frac{F_2}{A_2} \quad \text{or} \quad F_1\left(\frac{A_2}{A_1}\right) = F_2$$

4.6 Pascal's Principle

Simplified diagram of an automotive hydraulic brake system



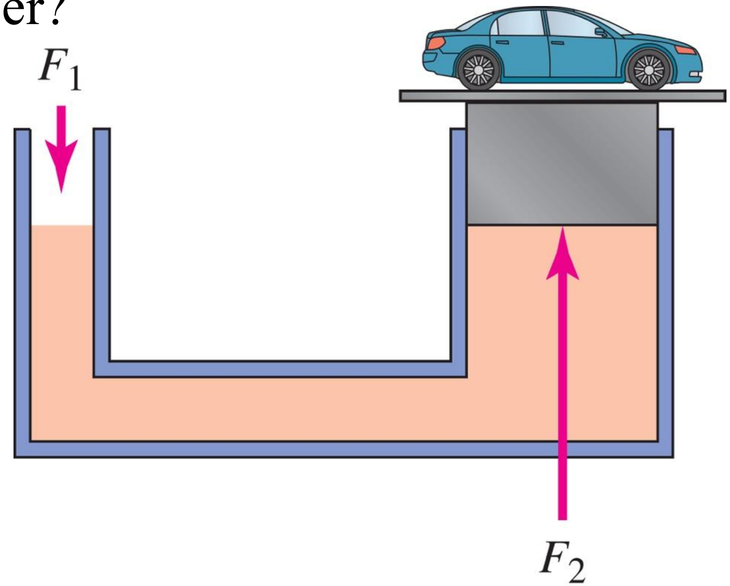
When the brake pedal is pushed, the pressure increase in the master cylinder is passed on through the fluid to the wheel cylinder. The piston is pushed outward, and the brake pads squeeze the disc.



(b)

Example 4.10

In a hydraulic car lift, the input piston has cross-sectional area $A_1 = 0.0025\ \text{m}^2$ and negligible weight. The output plunger has area $A_2 = 0.0625$ m2, and the weight of the car and the output plunger together is 17,500 N. If the input piston and output plunger are at the same level as shown in the figure, what is the size of the input force F_1 required to produce an output force F_2 capable of supporting the car and the output plunger?



$$p_1 = p_2 \rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$F_1 = F_2\left(\frac{A_1}{A_2}\right) = (17{,}500\ \text{N})\frac{(0.0025\ \text{m}^2)}{(0.0625\ \text{m}^2)} = 700\ \text{N}$$

4.7 Bernoulli's Principle

PRINCIPLES

Bernoulli's Principle: For a fluid undergoing steady flow, the pressure is lower where the fluid is flowing faster.

Steady flow means the fluid flow is smooth No random swirling or "white water"

4.7 Bernoulli's Principle

- This principle is based on the conservation of energy.

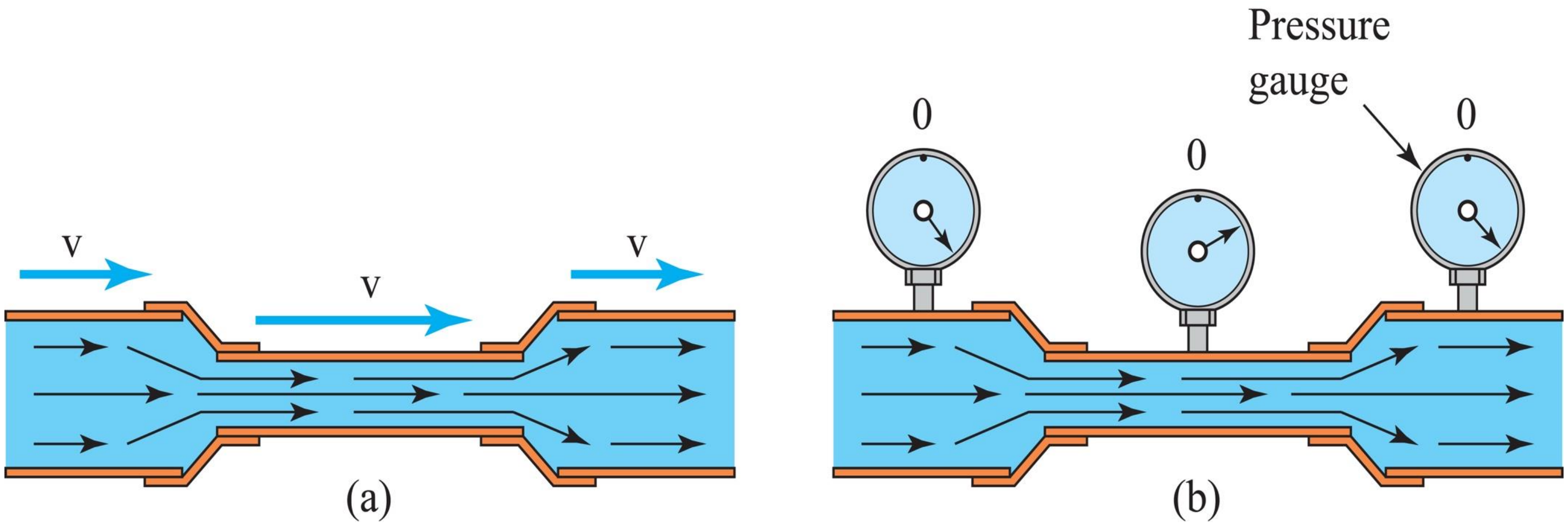
- A fluid under pressure has what can be called *pressure potential energy*.

  The higher the pressure, the greater the potential energy of any given volume of fluid.

- Moving fluids have both kinetic and potential energy.

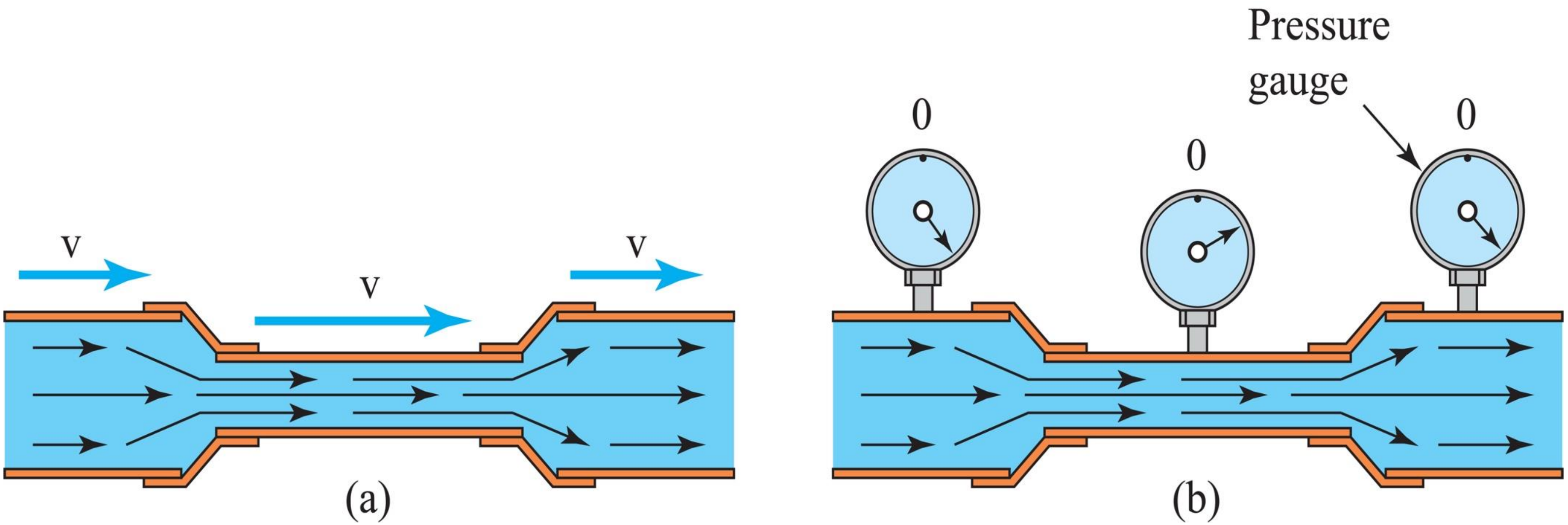
  When a fluid speeds up, its kinetic energy increases. Its total energy remains constant under the circumstances described, so its potential energy and, therefore, its pressure decrease.

4.7 Bernoulli's Principle



This is One of the best examples of Bernoulli's principle. Water flowing through a pipe passes through a smaller section spliced into the pipe. The water speeds up when it enters the narrow region and then slows down when it reenters the wide region.

4.7 Bernoulli's Principle



This occurs because the volume of fluid passing through each part of the pipe each second is the same. Where the cross-sectional area is smaller, the fluid must flow faster if the same number of cubic inches of fluid is to get through each second.

4.7 Bernoulli's Principle

You've seen this already if you've ever used your thumb to partially block water coming out of the end of a garden hose.

To quantify this relationship between flow velocity and cross-sectional area, we note that the *volume flow rate*, that is, the volume of fluid passing through a given cross-sectional area, A, of a horizontal pipe, is the product Av, where v is the fluid speed at that point along the pipe.

$$A_1 v_1 = A_2 v_2 \quad \text{(equation of continuity)}$$

4.7 Bernoulli's Principle

Under these circumstances, the volume flow rate is a constant of the fluid motion, and for any two points along the pipeline:

$$A_1 v_1 = A_2 v_2$$

Bernoulli's principle tells us that the pressure in the moving water is *smaller in the narrow section* than in the wide sections upstream and downstream.

Example 4.10

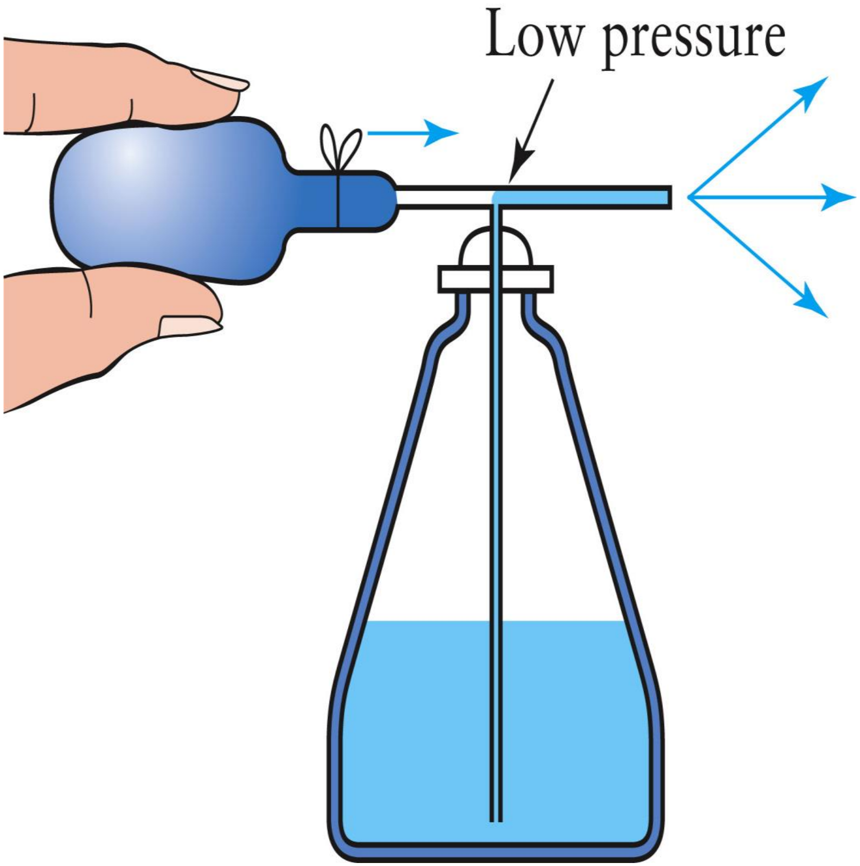
A common garden hose has an opening with a cross-sectional area of 5.1×10^{-4} m^2. When the spigot is opened, the water emerges from the hose with a speed of 0.85 m/s. If the gardener places her finger over the opening and reduces the area to 2.0×10^{-4} m^2, how fast will the water now exit the hose?

$$A_1 v_1 = A_2 v_2$$

$$v_2 = v_1 \left(\frac{A_1}{A_2} \right) = (0.85 \text{ m/s}) \frac{\left(5.1 \times 10^{-4} \text{ m}^2\right)}{\left(2.0 \times 10^{-4} \text{ m}^2\right)} = 2.17 \text{ m/s}$$

Bernoulli's Principle

Atomizers on perfume bottles utilize Bernoulli's principle. When a bulb is squeezed, air moves through a horizontal tube The air moves fast, so the air pressure is low. A second small tube runs from the horizontal tube down into the perfume. Because the air pressure is reduced in the horizontal tube, the normal air pressure acting on the surface of the perfume forces the liquid to rise upward in the vertical tube and to enter the moving air



Important Equations

IMPORTANT EQUATIONS

Equation	Comments	Equation	Comments
Fundamental Equations		*Special-Case Equations*	
$p = \frac{F}{A}$	Definition of pressure	$p = D_W h = Dgh$	Pressure (gauge) at a depth h in a liquid
$D = \frac{m}{V}$	Definition of mass density	$p = 0.433 \text{ psi/ft} \times h$	Pressure (gauge) in psi underwater at a depth h in feet
$D_W = \frac{W}{V}$	Definition of weight density	$A_1 v_1 = A_2 v_2$	Equation of continuity (constant density)
$F_b = W_{\text{fluid displaced}}$	Archimedes' principle	$\frac{F_1}{A_1} = \frac{F_2}{A_2}$	Hydraulic equation (constant density)